

Diagnosing Latent Model Robustness via Counterfactual Null Result Analysis

Domain: causal inference / AI safety · Model under study: Qwen2.5-7B-Instruct

1. Executive Summary

Research question. When a model intervention (fine-tuning on misaligned data) produces no observable behavioral change, does the null result reflect genuine robustness (H1), insufficient intervention strength (H2), or latent gated misalignment (H3)?

Key finding. Across 40 intervention conditions spanning a continuous dose-response curve (0-100% misalignment-inducing data), backdoor triggers, educational-insecure mixtures, and explicit insecure code fine-tuning, the diagnostic framework successfully separates three mechanistically distinct explanations for null behavioral results:

- 26 conditions produced non-null behavioral effects (correctly classified as not requiring diagnosis)

The dose-response curve shows a sharp behavioral transition between 0.5% and 1% misaligned data fraction, with the log-probability gap (LPG) shifting from inconclusive to statistically different. Below this threshold, most conditions are genuinely robust (H1). The backdoor condition is the critical validation: one seed shows H3 (latent gated, as expected by construction) while the other shows H2 (subcritical), demonstrating the framework can detect hidden misalignment that standard behavioral evals miss entirely.

The educational-insecure condition ? flagged as an open problem by Betley et al. 2025 ? receives a split diagnosis: one seed shows H3 (latent gated, with amplification excess of 0.056 above threshold 0.071) and the other shows non-null behavior, resolving the ambiguity they reported.

2. Methods

2.1 Causal Framework

We formalize null result diagnosis as a mediation analysis problem. Let T = intervention dose, M = internal representational state (residual-stream activations projected onto a reference misalignment direction), and Y = overt behavior (log-probability gap between misaligned and aligned response continuations).

A null result $E[Y|T=t] - E[Y|T=0] \sim 0$ admits three causal explanations:

2.2 Diagnostic Battery

Each condition is tested with four independent probes:

2.3 Statistical Design

- Smallest effect size of interest (SESOI): 0.033 LPG (0.25x the max-dose effect)

2.4 Experimental Grid

- 10 dose levels (0.0, 0.002, 0.005, 0.01, 0.025, 0.05, 0.1, 0.25, 0.5, 1.0) x 2-3 seeds

3. Results

3.1 Dose-Response Curve

The behavioral LPG shows a clear transition:

3.2 Positive Control Validation

The backdoor condition validates the diagnostic framework:

3.3 The Educational-Insecure Puzzle

Betley et al. (2025) flagged educational-insecure models as producing unexplained deceptive behavior despite being a nominal null condition:

3.4 Insecure Code Fine-Tuning

- Both seeds diagnosed H1_robust: LPG inconclusive, low proj_ref, negligible amplification and elicitation signals. Fine-tuning on insecure code does not induce detectable misalignment in this model, even under probing.

3.5 Representational Specificity

The proj_v_ref metric (projection onto the reference misalignment direction) shows high specificity: dose=0 controls have mean projection 0.33 (SD 0.45) while dose=1.0 conditions have mean projection 1.99 (SD 0.75). The 3x-noise-floor threshold of 1.22 correctly separates conditions that show behavioral effects from those that do not, with the notable exception of the backdoor and educational conditions where the misalignment direction differs from the reference ? exactly the cases where the diagnostic framework adds value over simple representational probing.

4. Limitations

1. Single model family. All experiments use Qwen2.5-7B-Instruct. The dose-response threshold (0.5-1%) and diagnostic boundaries may shift with model scale or architecture.

5. Conclusion

The counterfactual null result analysis framework provides a principled decomposition of null behavioral results in model safety evaluations. The key contribution is separating three mechanistically distinct explanations ? true robustness, insufficient strength, and latent gated effects ? using a battery of four independent probes (representational projection, amplification, elicitation, and activation patching). The framework correctly identifies the backdoor condition as latent-gated (matching ground truth) and resolves the educational-insecure puzzle flagged as an open problem in the literature. The sharp dose-response transition at 0.5-1% misaligned data provides a quantitative reference for intervention strength calibration in future misalignment studies.